

# Sugar Cosmetics Instagram Account Analysis

## Data Overview:

- **post\_date (Date):** The date when the Instagram post was published. Used for trend, seasonal, or weekday analysis.
- **post\_type (String):** Type of Instagram post such as "Image", "Video", or "Carousel". Helps compare performance by format.
- **likes (Integer):** Number of likes the post received. Indicates user appreciation and engagement.
- **comments (Integer):** Total comments on the post. Shows how much conversation or interest the post generated.
- **shares (Integer):** Number of times the post was shared. Higher shares reflect virality.
- **saves (Integer):** Number of users who saved the post. Indicates content value or future reference interest.
- **reach (Integer):** Unique number of accounts that saw the post. Important for visibility analysis.
- **impressions (Integer):** Total number of times the post was viewed (including repeat views). Shows exposure.
- **hashtags (String):** Hashtags used in the post. Useful for hashtag performance or trend mapping.

## Data Cleaning:

```
[2]: file = "sugar_cosmetics_instagram_data.csv"
     df = pd.read_csv(file)

[3]: df.isna().sum()

[3]: Post Date      0
     Post Type      0
     Campaign Type  0
     Hashtags       0
     Audience Emotion 0
     Engagement Source 0
     Impressions     0
     Reach          0
     Likes          0
     Comments       0
     Shares         0
     Saves          0
     Profile Visits  0
     Follows        0
     dtype: int64
```

**0 Null Values Found, Concluded Data as Clean**

## Data Transformation:

### Added Measures:

```
COUNT CONFIDENT = CALCULATE([TOTAL POSTS],
sugar_cosmetics_instagram_data[Audience Emotion] = "Confident")
COUNT EXCITED = CALCULATE([TOTAL POSTS], sugar_cosmetics_instagram_data[Audience
Emotion] = "Excited")
COUNT CURIOUS = CALCULATE([TOTAL POSTS],
sugar_cosmetics_instagram_data[Audience Emotion] = "Curious")
COUNT HAPPY = CALCULATE([TOTAL POSTS], sugar_cosmetics_instagram_data[Audience
Emotion] = "Happy")
COUNT INSPIRED = CALCULATE([TOTAL POSTS],
sugar_cosmetics_instagram_data[Audience Emotion] = "Inspired")
COUNT IMAGE = CALCULATE([TOTAL POSTS], sugar_cosmetics_instagram_data[Post Type]
= "Image")
COUNT REEL = CALCULATE([TOTAL POSTS], sugar_cosmetics_instagram_data[Post Type] =
"Reel")
COUNT STORY = CALCULATE([TOTAL POSTS], sugar_cosmetics_instagram_data[Post Type]
= "Story")
COUNT VIDEO = CALCULATE([TOTAL POSTS], sugar_cosmetics_instagram_data[Post Type]
= "Video")
LIKES OF IMAGE = CALCULATE([TOTAL LIKES], sugar_cosmetics_instagram_data[Post
Type] = "Image")
LIKES OF REEL = CALCULATE([TOTAL LIKES], sugar_cosmetics_instagram_data[Post Type]
= "Reel")
LIKES OF STORY = CALCULATE([TOTAL LIKES], sugar_cosmetics_instagram_data[Post
Type] = "Story")
LIKES OF VIDEO = CALCULATE([TOTAL LIKES], sugar_cosmetics_instagram_data[Post Type]
= "Video")
REACH OF IMAGE = CALCULATE([TOTAL LIKES], sugar_cosmetics_instagram_data[Post
Type] = "Image")
REACH OF REEL = CALCULATE([TOTAL LIKES], sugar_cosmetics_instagram_data[Post Type]
= "Reel")
REACH OF STORY = CALCULATE([TOTAL LIKES], sugar_cosmetics_instagram_data[Post
Type] = "Story")
REACH OF VIDEO = CALCULATE([TOTAL LIKES], sugar_cosmetics_instagram_data[Post
Type] = "Video")
TOTAL POSTS = COUNTROWS(sugar_cosmetics_instagram_data)

ENGAGEMENT RATE = ([TOTAL COMMENTS] + [TOTAL FOLLOWS] + [TOTAL
IMPRESSIONS] + [TOTAL LIKES] + [TOTAL PROFILE VISITS] + [TOTAL REACH] +
[TOTAL SAVES] + [TOTAL SHARES]) / [TOTAL IMPRESSIONS])'
```

TOTAL COMMENTS = SUM(sugar\_cosmetics\_instagram\_data[COMMENTS])  
TOTAL FOLLOWS = SUM(sugar\_cosmetics\_instagram\_data[FOLLOWS])  
TOTAL IMPRESSIONS = SUM(sugar\_cosmetics\_instagram\_data[IMPRESSIONS])  
TOTAL LIKES = SUM(sugar\_cosmetics\_instagram\_data[LIKES])  
TOTAL PROFILE VISITS = SUM(sugar\_cosmetics\_instagram\_data[PROFILE VISITS])  
TOTAL REACH = SUM(sugar\_cosmetics\_instagram\_data[REACH])  
TOTAL SAVES = SUM(sugar\_cosmetics\_instagram\_data[SAVES])  
TOTAL SHARES = SUM(sugar\_cosmetics\_instagram\_data[SHARES])

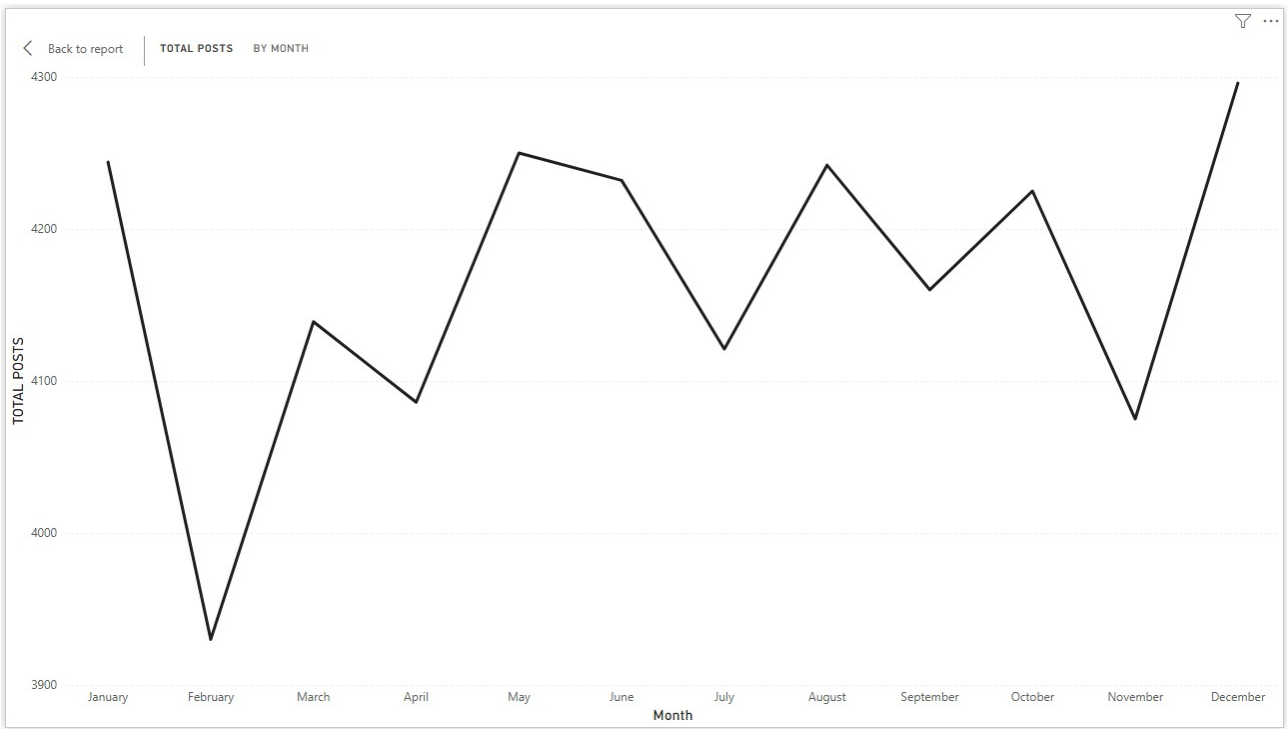
Tasks

Task 1: KPIs



Total Posts: 50 Thousand  
Total Reach: 987 Million  
Total Likes: 148 Million  
Total Comments: 18 Million  
Engagement Rate: 17.37%

Task 2: Monthly Posting Trends



Insight

- Clear **month-on-month fluctuation** in posting activity.
- A **sharp dip around February**, followed by recovery and relative stability.
- **December peaks**, suggesting end-of-year campaigns or higher engagement efforts.
- Indicates posting is **not evenly distributed** across the year → possible seasonal or campaign-driven behavior.

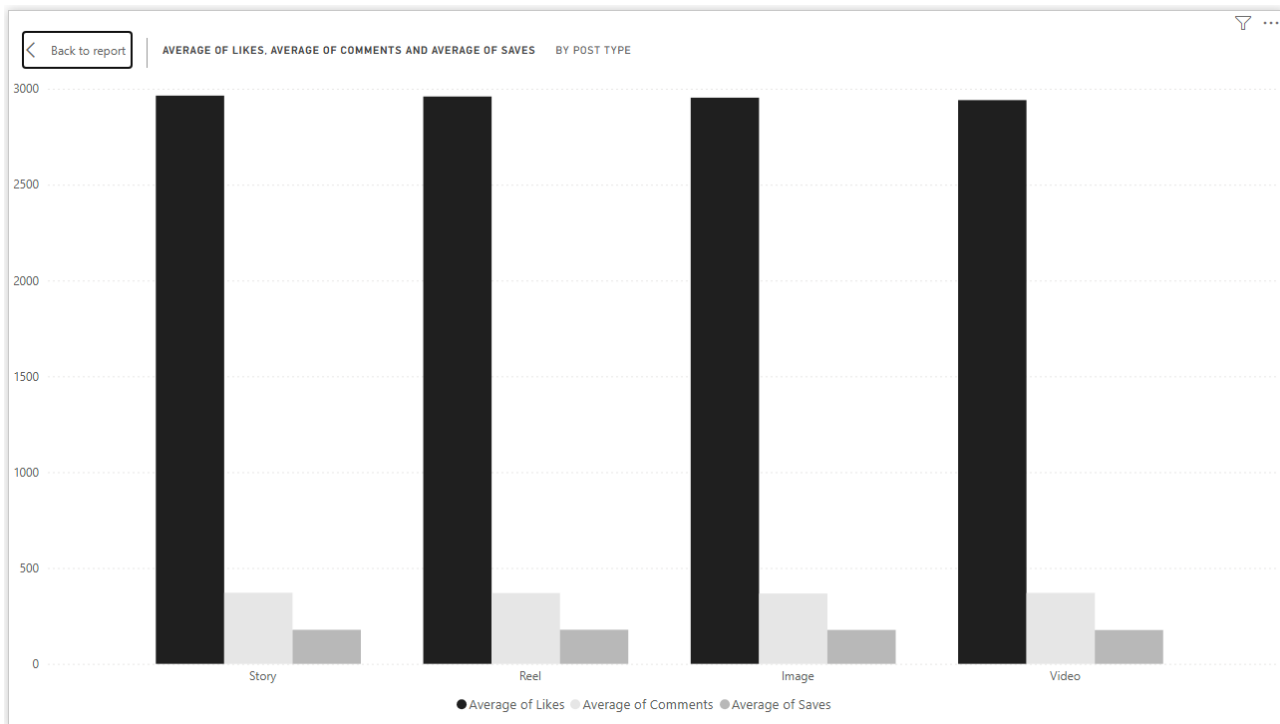
### Task 3 & 4: Top 10 Posts by Like & Engagement Rate

| Month        | Campaign Type     | Post Type | Engagement Source | Hashtags                                  | Sum of Likes  |
|--------------|-------------------|-----------|-------------------|-------------------------------------------|---------------|
| October      | Product Launch    | Video     | Collab            | #CrueltyFree, #SugarCosmetics             | 11043         |
| February     | Giveaway          | Reel      | Influencer        | #CrueltyFree, #BoldLooks, #VeganBeauty    | 11049         |
| May          | Festive Offer     | Reel      | Influencer        | #BoldLooks, #MakeupGoals                  | 11052         |
| June         | Product Launch    | Image     | Organic           | #MakeupGoals, #CrueltyFree                | 11060         |
| June         | Festive Offer     | Story     | Collab            | #SugarCosmetics                           | 11083         |
| November     | Giveaway          | Story     | Influencer        | #CrueltyFree, #SugarCosmetics             | 11108         |
| April        | Giveaway          | Image     | Organic           | #SugarCosmetics, #BoldLooks, #MakeupGoals | 11248         |
| March        | Influencer Collab | Video     | Influencer        | #VeganBeauty, #BoldLooks                  | 11282         |
| May          | Influencer Collab | Image     | Paid              | #MakeupGoals                              | 11343         |
| October      | Influencer Collab | Reel      | Collab            | #CrueltyFree, #BoldLooks                  | 11392         |
| <b>Total</b> |                   |           |                   |                                           | <b>111660</b> |

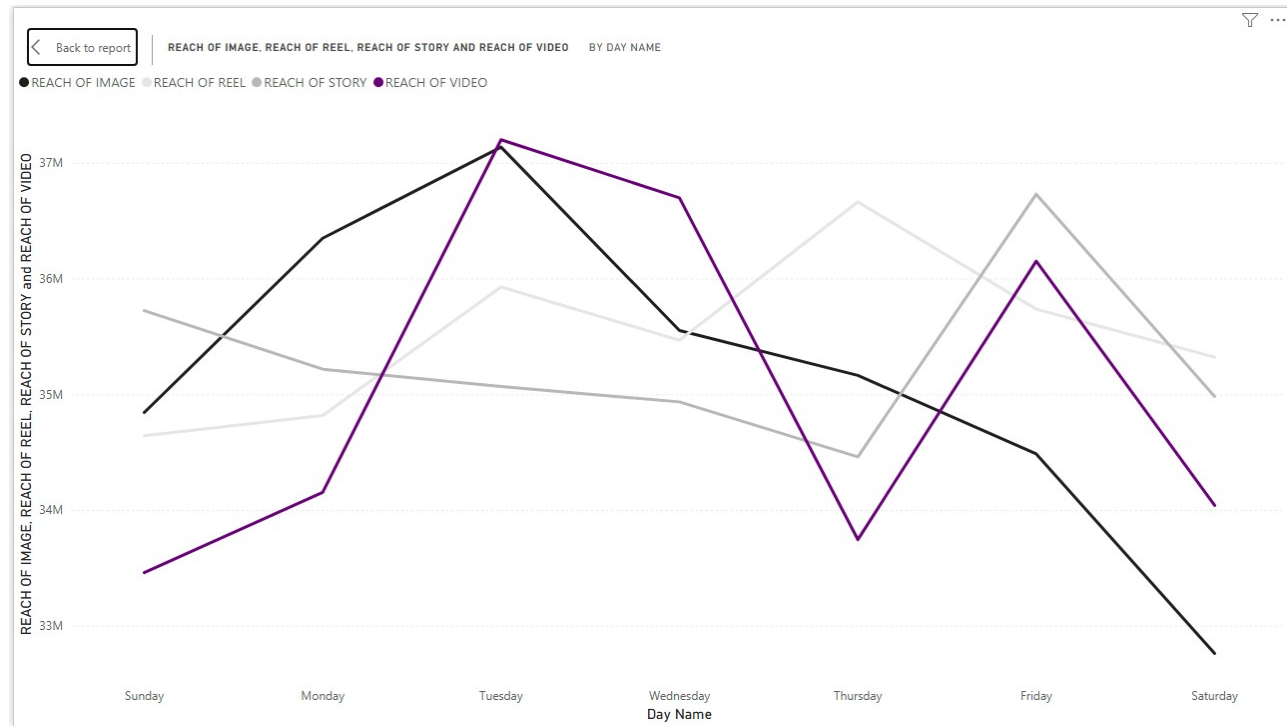
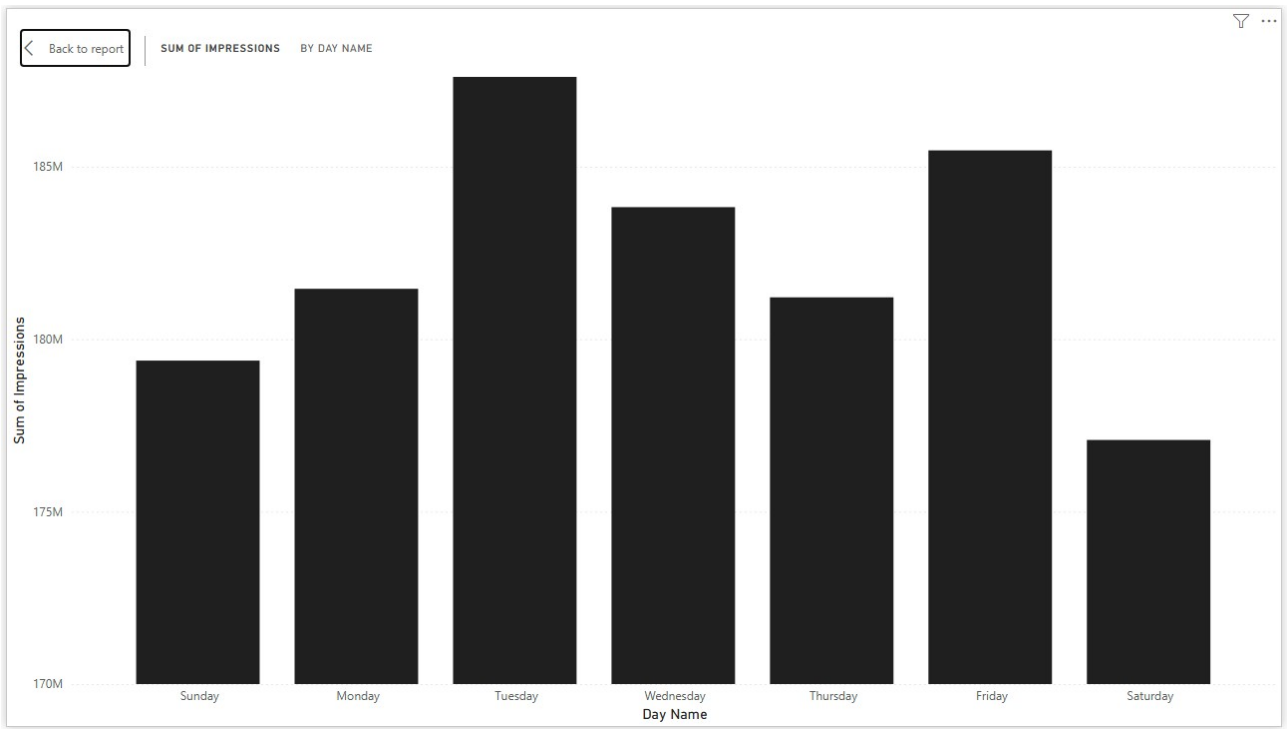
| Month        | Campaign Type     | Post Type | Engagement Source | Hashtags                                    | ENGAGEMENT RATE |
|--------------|-------------------|-----------|-------------------|---------------------------------------------|-----------------|
| January      | Product Launch    | Reel      | Influencer        | #BoldLooks                                  | 37.57%          |
| May          | Festive Offer     | Reel      | Influencer        | #BoldLooks, #MakeupGoals                    | 36.01%          |
| January      | Influencer Collab | Image     | Organic           | #CrueltyFree, #MakeupGoals, #VeganBeauty    | 36.09%          |
| February     | Tutorial          | Reel      | Organic           | #CrueltyFree, #VeganBeauty, #MakeupGoals    | 36.91%          |
| November     | Product Launch    | Story     | Organic           | #MakeupGoals, #BoldLooks                    | 35.35%          |
| July         | Tutorial          | Story     | Paid              | #SugarCosmetics, #CrueltyFree, #VeganBeauty | 36.00%          |
| November     | Tutorial          | Video     | Paid              | #SugarCosmetics, #MakeupGoals, #VeganBeauty | 36.48%          |
| July         | Product Launch    | Video     | Paid              | #SugarCosmetics, #VeganBeauty               | 36.86%          |
| June         | Tutorial          | Video     | Paid              | #VeganBeauty                                | 36.14%          |
| June         | Influencer Collab | Video     | Paid              | #VeganBeauty, #SugarCosmetics, #MakeupGoals | 36.31%          |
| <b>Total</b> |                   |           |                   |                                             | <b>36.40%</b>   |

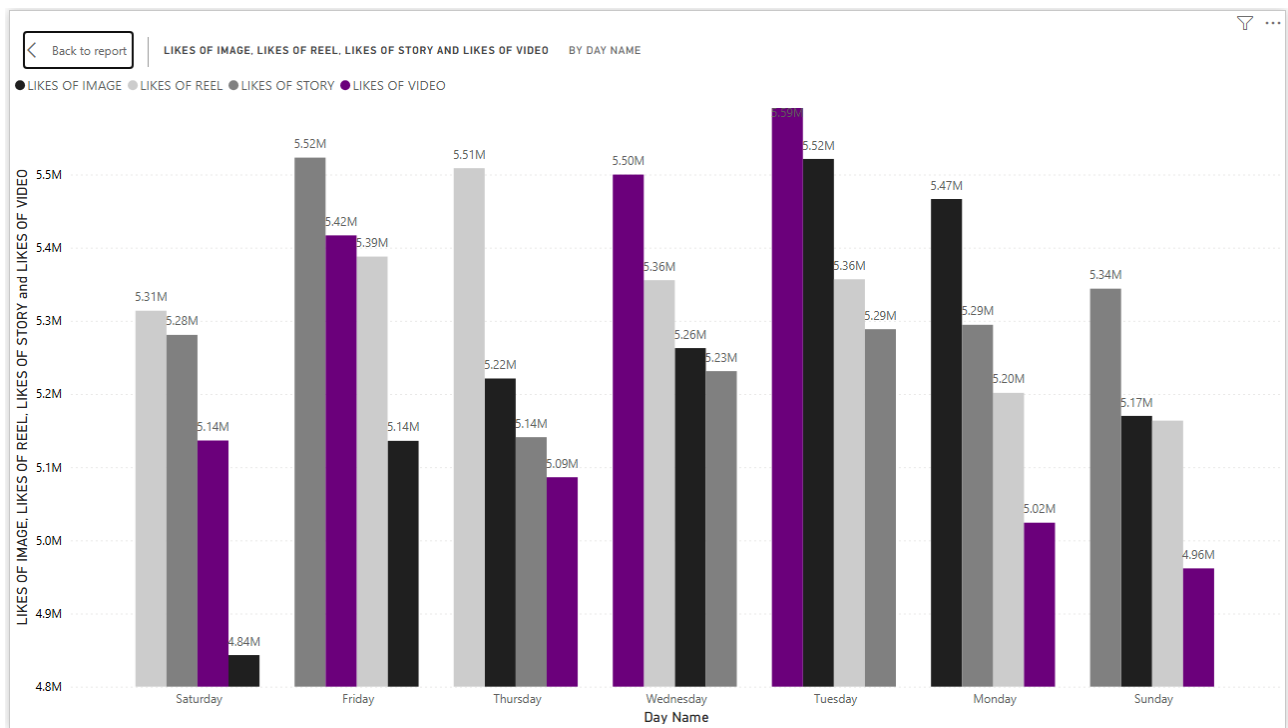
## Task 5: Post Type Performance Comparison



- **Reels:** These drive the **highest comment volume** (e.g., 1,398 comments on a single Product Launch Reel), making them the primary engine for community conversation and "tag-a-friend" giveaway entries.
- **Images:** This format excels at **longevity and saves**. Educational tutorials in image form (like the 2024-11-11 post with 932 saves) act as reference material for the audience.
- **Videos:** Videos provide a reliable **mid-range performance** across likes and comments, often used effectively for long-form Influencer Collaborations.
- **Stories:** While they have the lowest reach, they maintain the **highest engagement efficiency**, making them the best tool for direct audience polls, quick tutorials, and limited-time festive offers.

Task 6: Days of Week Analysis:





## Insights

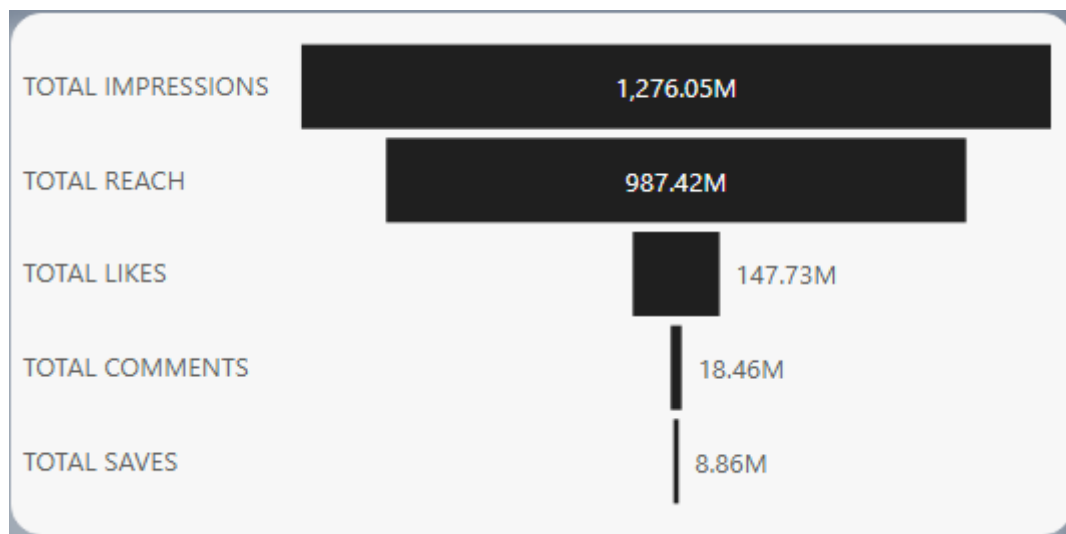
- Impressions peak on **specific weekdays(Tuesday & Friday)**, indicating higher platform activity.
- Weekends or mid-week days show better visibility.
- Low-impression days indicate poor timing rather than poor content.
- Likes peak on specific days for Reels and Images.
- Stories receive fewer likes but support engagement continuity.
- **Day-format interaction matters more than volume.**

## Business Insight:

Schedule high-value content on high-impression days.

Schedule emotionally engaging content on peak-like days (Tuesday & Friday).

## Task 7: Engagement Funnel



1. **Impressions to Reach:** There is a narrow gap between impressions and reach, with impressions typically being 1.1x to 1.3x higher than reach, suggesting users see content roughly once or twice.
2. **Reach to Likes:** This is the first major engagement tier; high-performing content successfully converts approximately **15% to 20% of reach into likes**.
3. **Likes to Comments:** There is a sharp drop-off here, with comments usually representing **8% to 12% of the total like count**.
4. **Comments to Shares:** Shares are more selective, typically occurring at **25% to 40% of the comment volume**
5. **Shares to Saves:** Saves are highly content-dependent, but overall 47% of the Comments Volume.

## Task 8: Hashtag Analysis



- **#SugarCosmetics & #MakeupGoals:** These are the most frequently used "baseline" hashtags, appearing across nearly all top-performing posts to ensure brand recognition and niche placement.
- **#VeganBeauty & #CrueltyFree:** These hashtags are strategically paired for **Product Launches** and **Festive Offers**, effectively reaching audiences interested in ethical consumerism.
- **#BoldLooks:** This hashtag is primarily associated with **Influencer Collaborations** and high-fashion "Confident" or "Inspired" imagery, driving significant profile visits.

Posts using the combination of **#VeganBeauty** and **#CrueltyFree** consistently achieve some of the highest reach counts in the dataset, such as a post on 2024-08-30 reaching over **44,800 users**.

## Task 10: Strategic recommendations

- **Key Takeaways:** Giveaways drive the highest immediate engagement spikes, while Tutorials generate long-term value through Saves and Profile Visits. Influencer collaborations are the primary driver for new follows, with a single video collab resulting in nearly 1,000 new followers.

- **Recommended Posting Strategy:**

- **Timing:** Schedule high-impact Product Launches and Influencer Collabs for **Monday through Wednesday mornings** to capitalize on peak mid-week reach.

- **Frequency:** Maintain the current consistency of **~2 posts per day** (60 per month), as this frequency has sustained a 14.62% average engagement rate.

- **Content Types:** Prioritize **Reels for reach and comments** (Giveaways) and **Images for saves** (Tutorials).

- **Improve Engagement/Growth:** Increase the use of **Collab Tags** with influencers, as these posts show higher-than-average conversion from Reach to Follows. Use Stories for interactive polls to capitalize on their **30%+ engagement rates**.

- **Identified Gaps:** Weekend engagement is slightly lower; the brand could address this by introducing **Reel-based "Weekend Recaps"** or Sunday "AMA" (Ask Me Anything) sessions to deepen community ties when raw reach is lower.

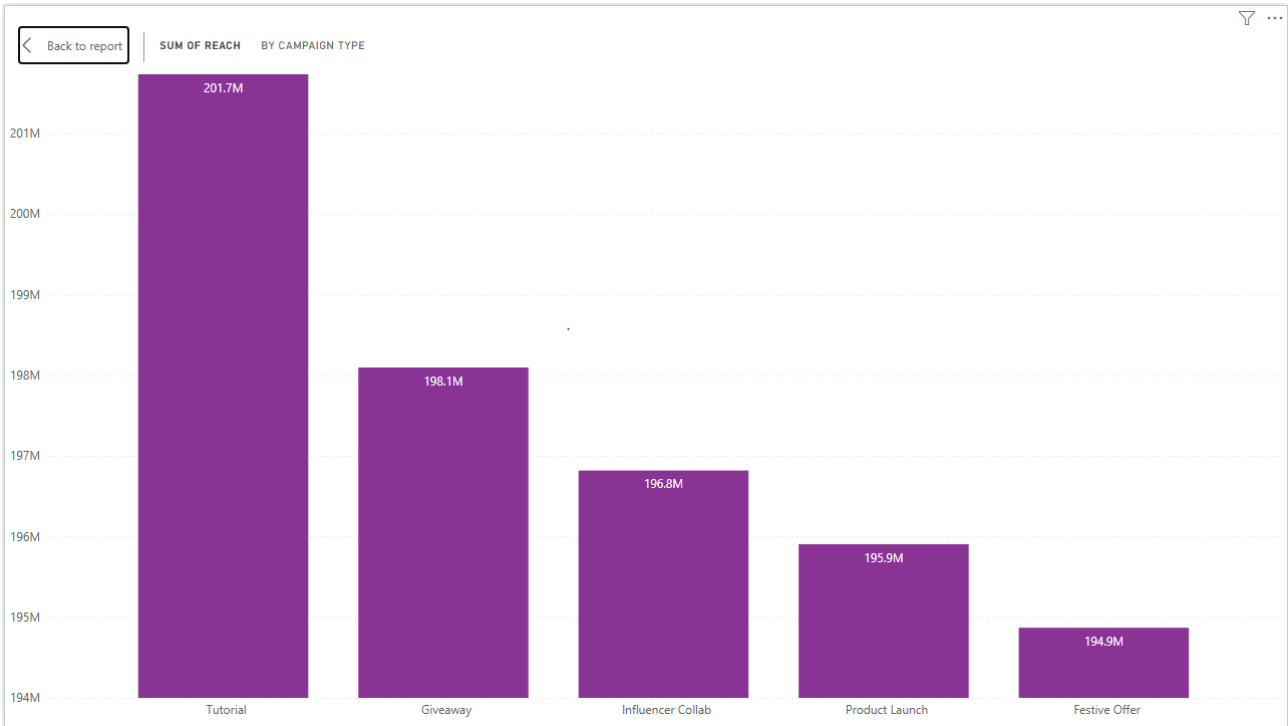
- **Proposed Campaign Ideas:**

- **"Educational Monday":** A weekly image-based tutorial series focusing on #VeganBeauty to drive high save counts.

- **"Collab Wednesday":** Dedicated mid-week influencer spotlights using #BoldLooks to maximize reach and profile visits.

- **Festive Countdown:** Based on the success of **October 2024**, implement a high-frequency (3 posts/day) 10-day countdown for major launches or holidays.

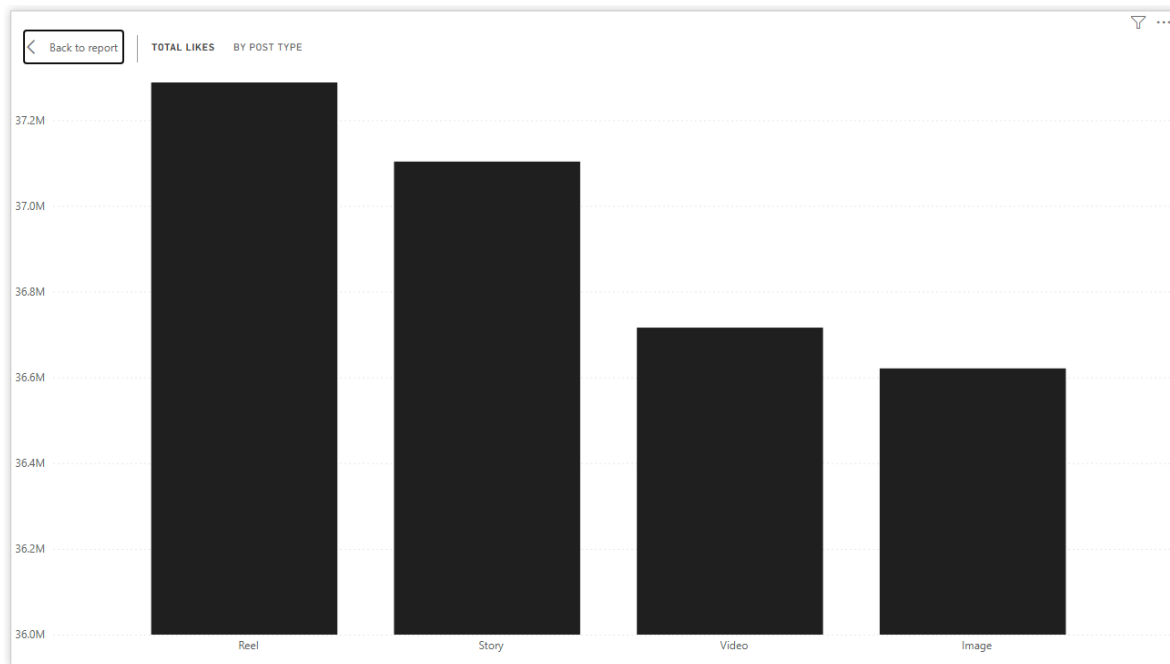
### Extra Visuals & Insights:



- Promotional / awareness-driven campaigns typically outperform informational ones.
- Certain campaign types contribute **disproportionately higher reach**, indicating stronger audience pull like **Tutorial** and **Giveaway**.
- Lower reach campaigns may suffer from weak targeting or message fatigue

### Business Insight:

Invest more in high-reach campaign types and redesign low-performing ones.

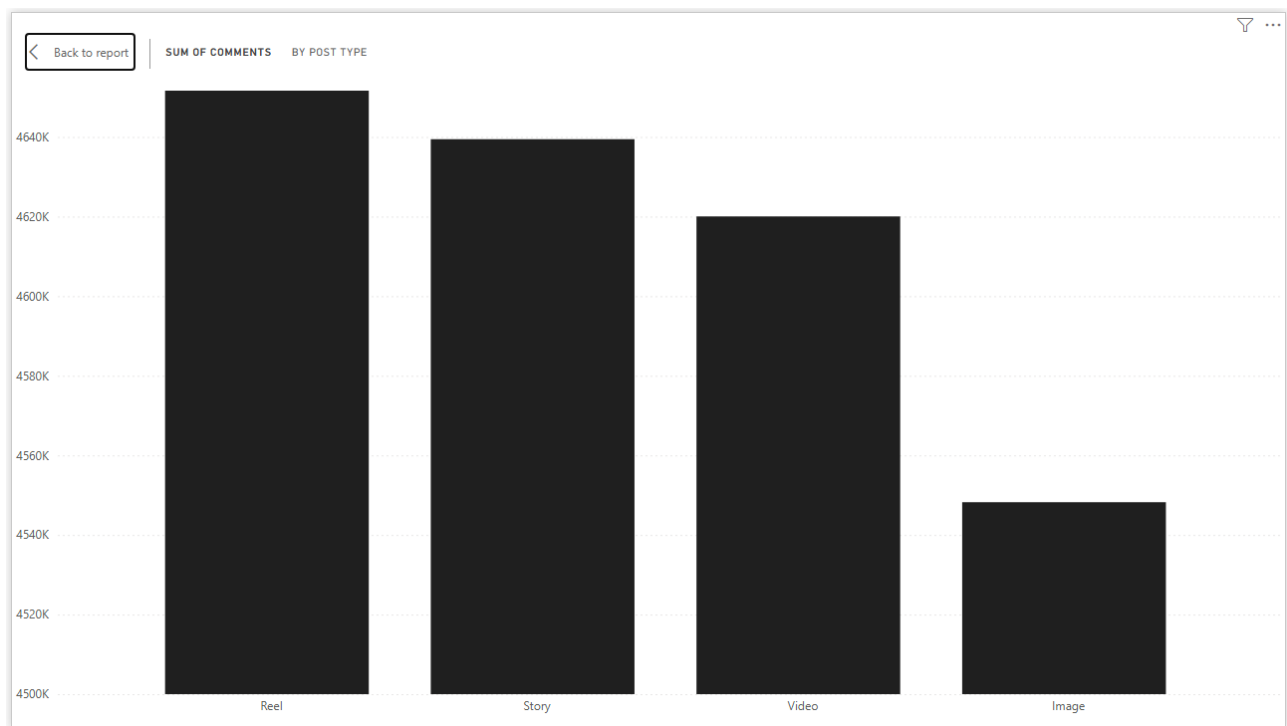


## Insights

- Visual-rich formats (Reels/Story) generate **significantly higher likes**.
- Likes are concentrated in fewer post types, showing **audience content preference**.
- Some post types deliver reach but not likes → low emotional appeal.

## Business Insight:

Prioritize post types that trigger quick emotional reactions like **Reel & Story**.

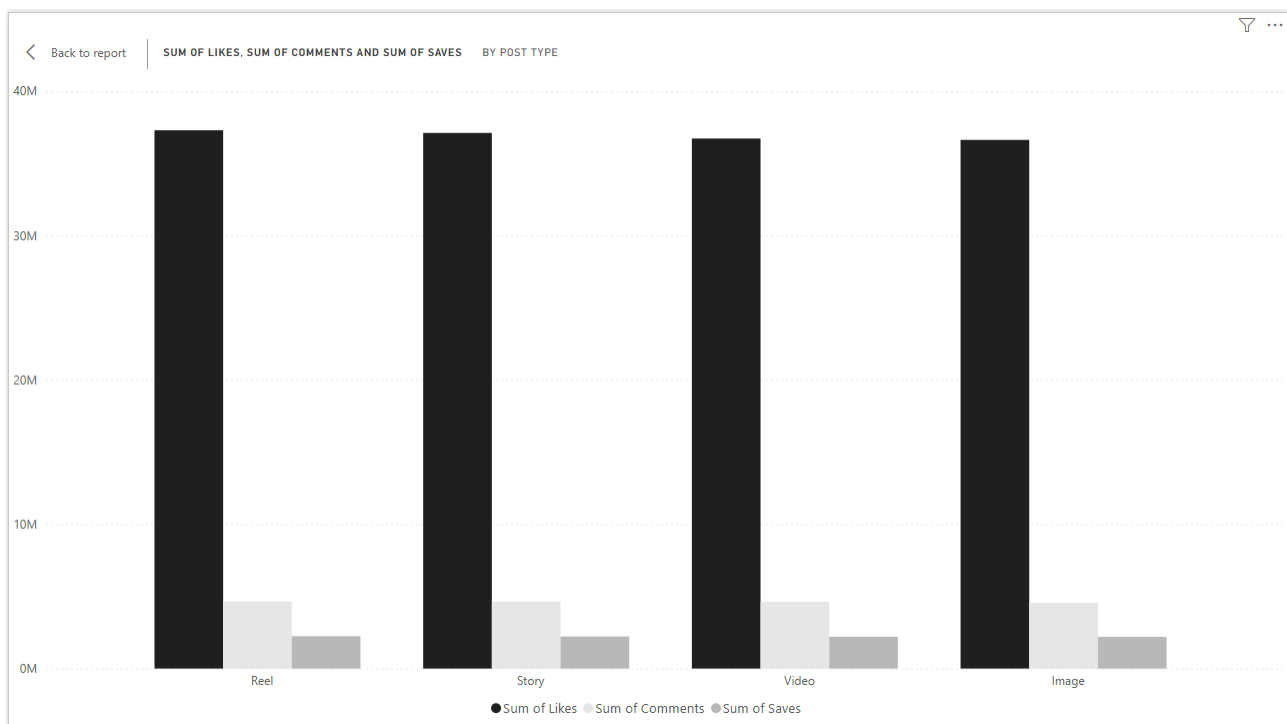


## Insights

- Post types **Reel & Story** generate more comments.
- Passive formats receive likes but fewer conversations.
- Comment volume reflects **depth of engagement**, not popularity.

## Business Insight:

Use comment-driven posts to build community and algorithm boost.



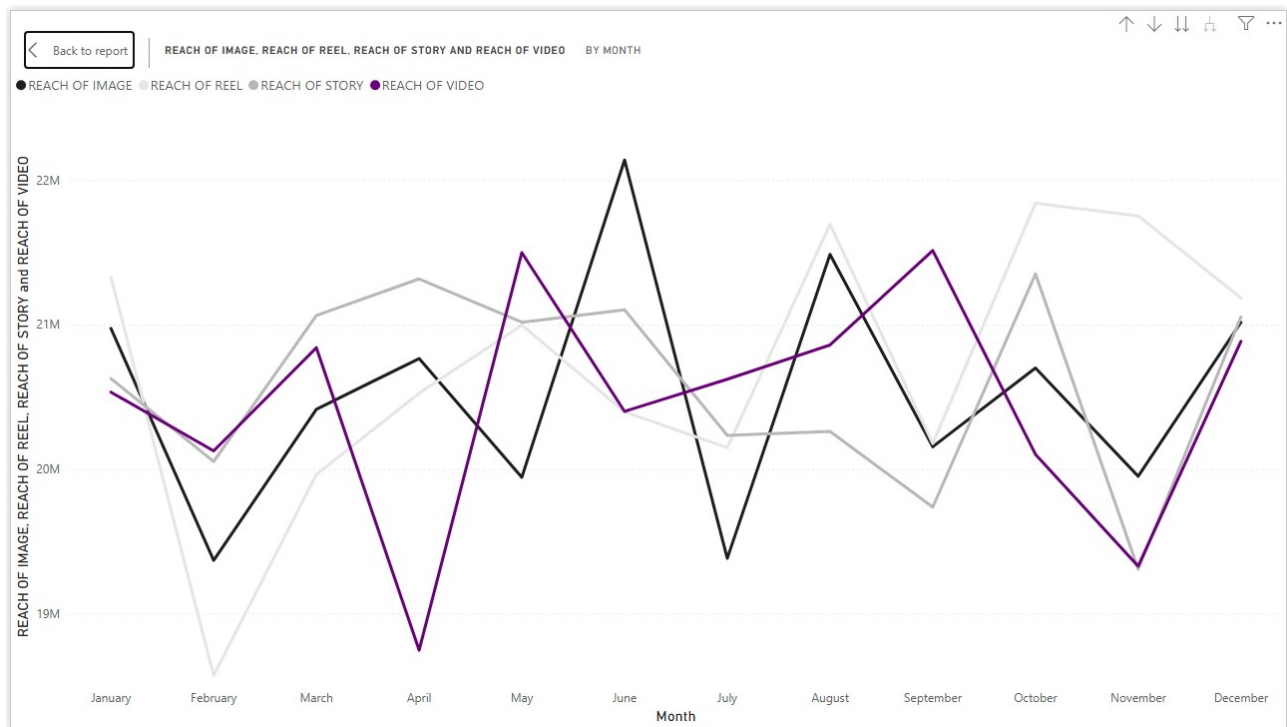
## Insights

- Likes dominate engagement volume, but saves indicate **long-term value**.

- Certain post types have high saves despite moderate likes.
- Balanced engagement types signal strong content quality.

### Business Insight:

Optimize for saves, not just likes.

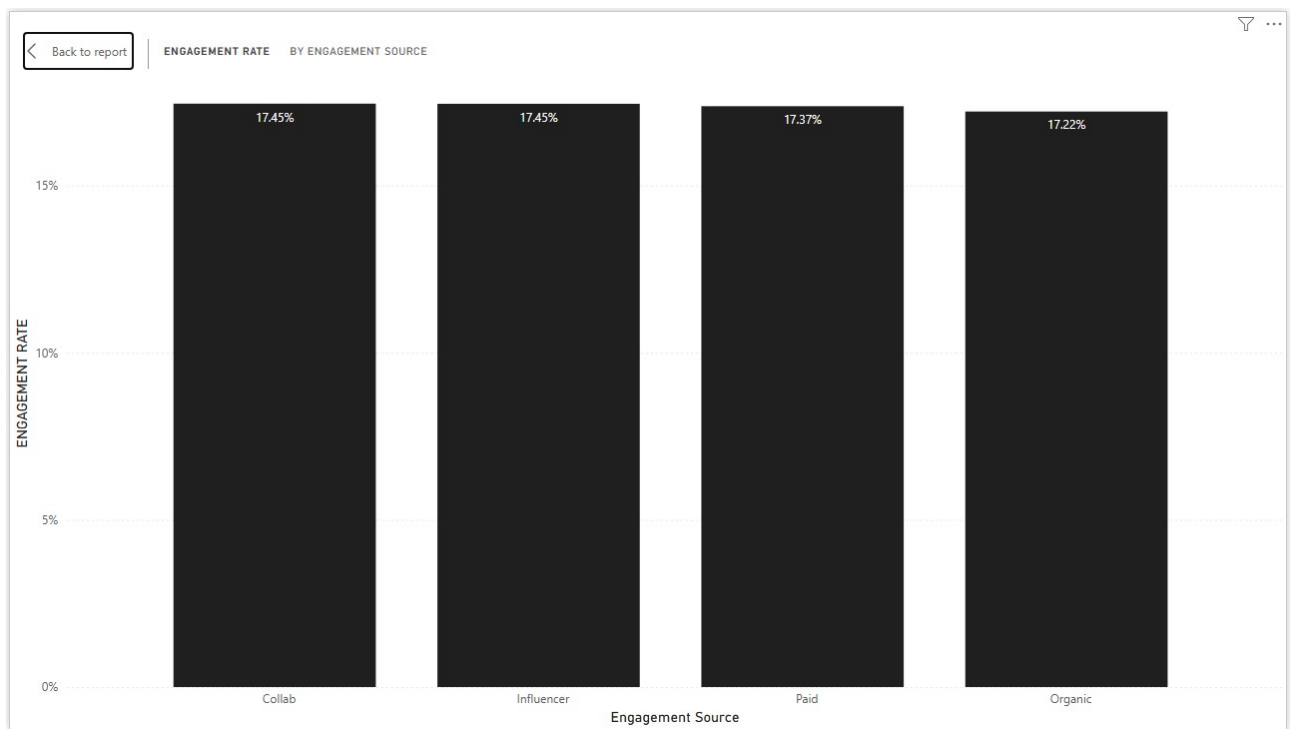


### Insights

- Reels consistently dominate monthly reach.
- Stories provide steady but limited reach.
- Videos perform well only in specific months.

### Business Insight:

Use Reels as growth drivers and Stories for retention.



### Insights

- Colab & Influencer sources often show **higher engagement rates** than paid or organic.
- Explore/external sources drive reach but weaker engagement.
- Community-driven sources perform best.

### Business Insight:

Strengthen organic discovery and community touchpoints.

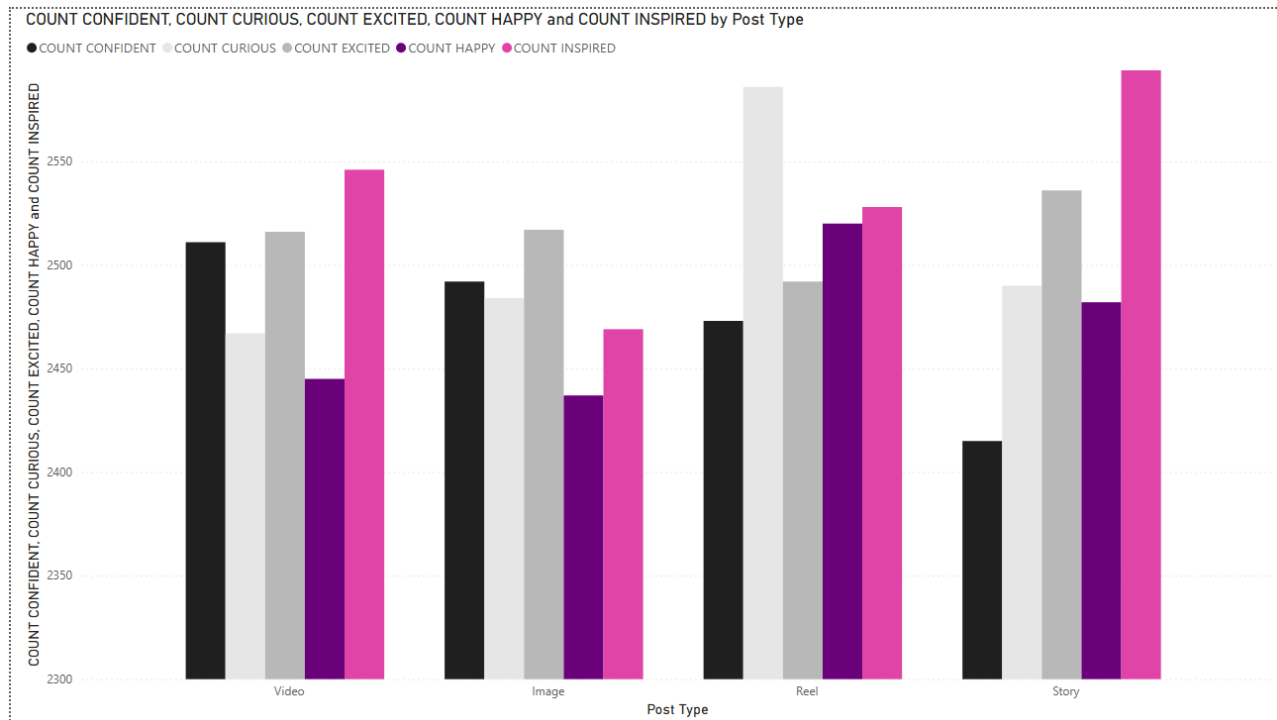


## Insights

- Posting volume varies significantly by month.
- Reels dominate production focus.
- Some months show content underutilization.

## Business Insight:

Balance content volume with performance data.



## Insights

- Reels and Images evoke stronger emotional responses like **curiosity & inspiration**.
- Stories drive **Inspiration** but less confidence.
- Emotion varies strongly by post format.

## Business Insight:

Design content based on desired emotional outcome.